SQL window functions, finally simple

Rank rows

Who is first, second, third

Data: Practice tables (example): employees (8 staff, 3 depts, salary), sales (North and South, Jan to Mar 2024), orders (8 orders, 4 customers).

1 Number every row

→ Asha 1, Ravi 2, Karan 3, ...

```
SELECT name, ROW_NUMBER() OVER (ORDER BY
salary DESC) FROM employees;
```

2 Rank, ties share a place

→ Asha 1, Ravi 2, Karan 2, ...

```
SELECT name, RANK() OVER (ORDER BY salary
DESC) FROM employees;
```

3 Rank with no gaps

→ Asha 1, Ravi 2, Karan 2, ...

```
SELECT name, DENSE_RANK() OVER (ORDER BY
salary DESC) FROM employees;
```

SQL WINDOWS · 2/2

Rank rows

Who is first, second, third

4 Rank inside each dept

→ HR Neha 1, HR Sneha 2, IT Asha 1, ...

```
SELECT dept, name, RANK() OVER (PARTITION BY
dept ORDER BY salary DESC) FROM employees;
```

5 Top earner per dept

→ HR Neha, IT Asha, Sales Vikas

```
WITH r AS (SELECT name, dept, RANK() OVER
(PARTITION BY dept ORDER BY salary DESC) AS
rk FROM employees) SELECT dept, name FROM r
WHERE rk = 1;
```

Interview trap: Ravi and Karan both earn 61,000. RANK gives them 2 and 2, then jumps to 4.

SQL WINDOWS · 1/2

Running totals

Totals without losing your rows

1 Running total by region

```
→ North 2024-01 1,000, North 2024-02 2,200, North 2024-03 3,100, ...
```

```
SELECT region, month, SUM(sales) OVER (PARTITION BY region ORDER BY month) FROM sales;
```

2 Region total on every row

```
→ North 2024-01 3,100, North 2024-02 3,100, North 2024-03 3,100, ...
```

```
SELECT region, month, SUM(sales) OVER (PARTITION BY region) FROM sales;
```

3 Share of region, %

```
→ North 2024-01 32.3, North 2024-02 38.7, North 2024-03 29, ...
```

```
SELECT region, month, ROUND(100.0 * sales / SUM(sales) OVER (PARTITION BY region), 1) FROM sales;
```

SQL WINDOWS · 2/2

Running totals

Totals without losing your rows

4 3-month moving average

```
→ North 2024-01 1,000, North 2024-02 1,100, North 2024-03 1,033, ...
```

```
SELECT region, month, ROUND(AVG(sales) OVER (PARTITION BY region ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)) FROM sales;
```

5 Dept average beside salary

```
→ Asha 72,000 64,667, Ravi 61,000 64,667, Neha 48,000 46,500, ...
```

```
SELECT name, salary, ROUND(AVG(salary) OVER (PARTITION BY dept)) FROM employees ORDER BY id;
```

Remember: GROUP BY squashes rows. OVER keeps every row and adds the total beside it.

SQL WINDOWS · 1/2

Compare with last month

LAG and LEAD do the VLOOKUP-to-row-above for you

1 Previous month's sales

```
→ North 2024-01 NULL, North 2024-02 1,000, North
2024-03 1,200, ...
```

```
SELECT region, month, LAG(sales) OVER
(PARTITION BY region ORDER BY month) FROM
sales;
```

2 Month-on-month change

```
→ North 2024-01 NULL, North 2024-02 200, North
2024-03 -300, ...
```

```
SELECT region, month, sales - LAG(sales) OVER
(PARTITION BY region ORDER BY month) FROM
sales;
```

3 Next month's sales

```
→ North 2024-01 1,200, North 2024-02 900, North
2024-03 NULL, ...
```

```
SELECT region, month, LEAD(sales) OVER
(PARTITION BY region ORDER BY month) FROM
sales;
```

Compare with last month

LAG and LEAD do the VLOOKUP-to-row-above for you

4 First month as baseline

```
→ North 2024-01 1,000, North 2024-02 1,000, North 2024-03 1,000, ...
```

```
SELECT region, month, FIRST_VALUE(sales)
OVER (PARTITION BY region ORDER BY month)
FROM sales;
```

5 Split staff into 4 pay bands

```
→ Asha 1, Ravi 1, Karan 2, ...
```

```
SELECT name, NTILE(4) OVER (ORDER BY salary
DESC) FROM employees;
```

Say this: LAG looks back, LEAD looks ahead. The first row has nothing behind it, so you get NULL.

SQL WINDOWS

Interview classics

Asked in almost every SQL round

1 Second highest salary

→ 61,000

```
SELECT DISTINCT salary FROM (SELECT salary,
DENSE_RANK() OVER (ORDER BY salary DESC) AS dr
FROM employees) WHERE dr = 2;
```

2 Find duplicate emails

→ asha@corp.in

```
SELECT email FROM (SELECT email, ROW_NUMBER()
OVER (PARTITION BY email ORDER BY id) AS rn
FROM employees) WHERE rn > 1;
```

3 Latest order per customer

→ 1 2024-03-15, 2 2024-04-02, 3 2024-03-01, ...

```
SELECT customer_id, order_date FROM (SELECT
customer_id, order_date, ROW_NUMBER() OVER
(PARTITION BY customer_id ORDER BY order_date
DESC) AS rn FROM orders) WHERE rn = 1;
```

Interview trap: You cannot use a window function in WHERE. Wrap it in a CTE or subquery first.

SQL WINDOWS

Which one do I use?

You need → Use

1. **A unique number on every row**
   ROW_NUMBER
2. **Ties to share a rank, gaps after**
   RANK
3. **Ties to share a rank, no gaps**
   DENSE_RANK
4. **The row above or below**
   LAG / LEAD
5. **A running or moving total**
   SUM / AVG with ORDER BY inside OVER
6. **A group total beside each row**
   SUM with PARTITION BY, no ORDER BY
7. **Top N in every group**
   ROW_NUMBER or RANK, then filter in a CTE
8. **Equal-sized buckets**
   NTILE

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